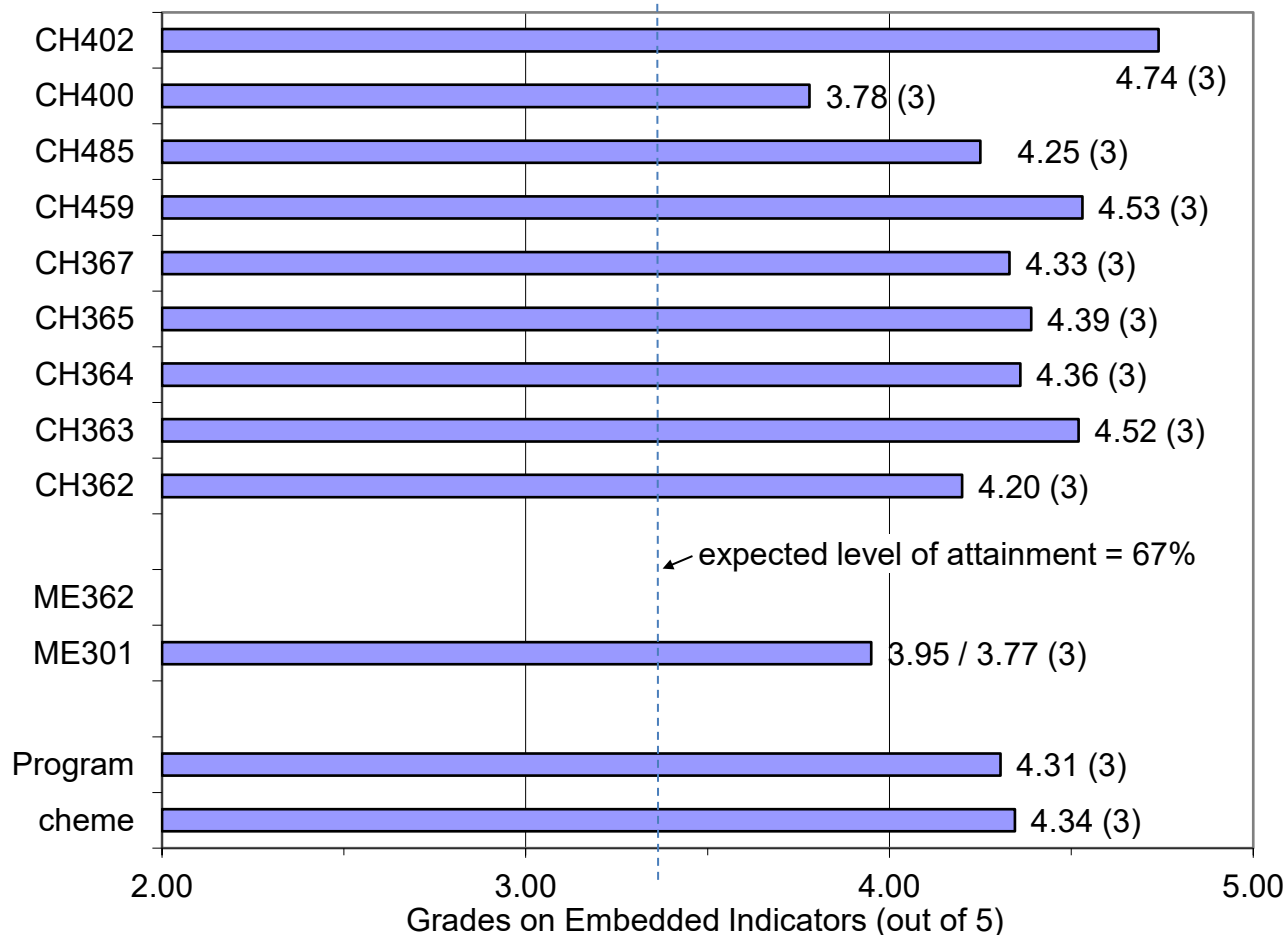




Example Data: Coursework Embedded Indicators

Student Outcome 1 – Example Data

Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.



Values in parentheses are coverage ratings from Table 5-4 in the 2020 Self-Study and explained on page 5-7.

Parentheses ratings:

3: Unique embedded indicator with clear rubric or cut scale.

2: Outcome was graded but grades are convoluted, or part of the outcome is not covered.

1: Correlation to outcome but no assessment

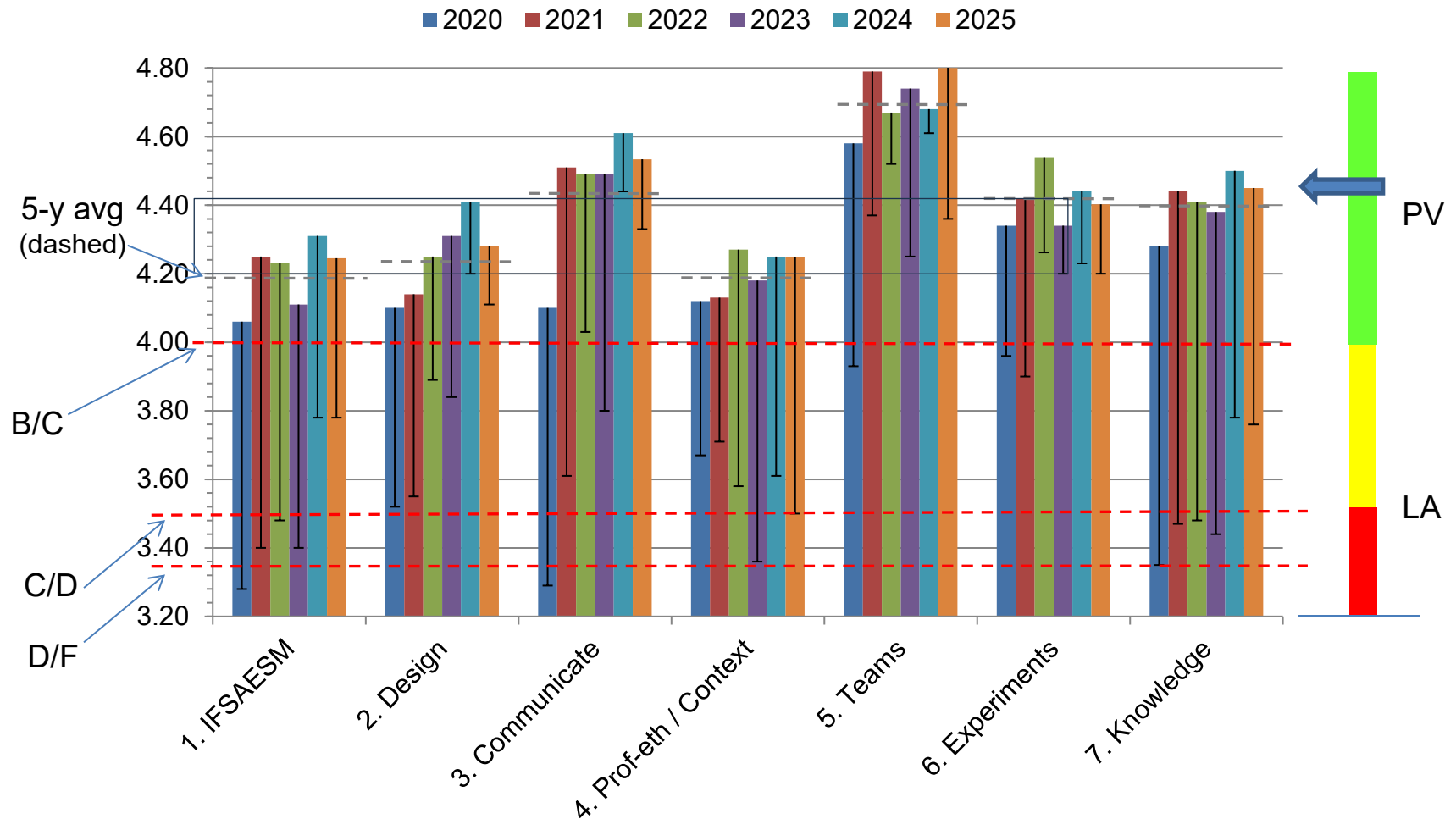
0: No coverage or correlation

Data shown here is for Class of 2025
Similar data is collected for all 7 ABET student outcomes
Summary of all data is shown on next slide



Performance on Embedded Indicators

Program Averages AY2020 to AY2025



All indicators are in the green and trending upward.

Error bars are minimum scores from course assessments.

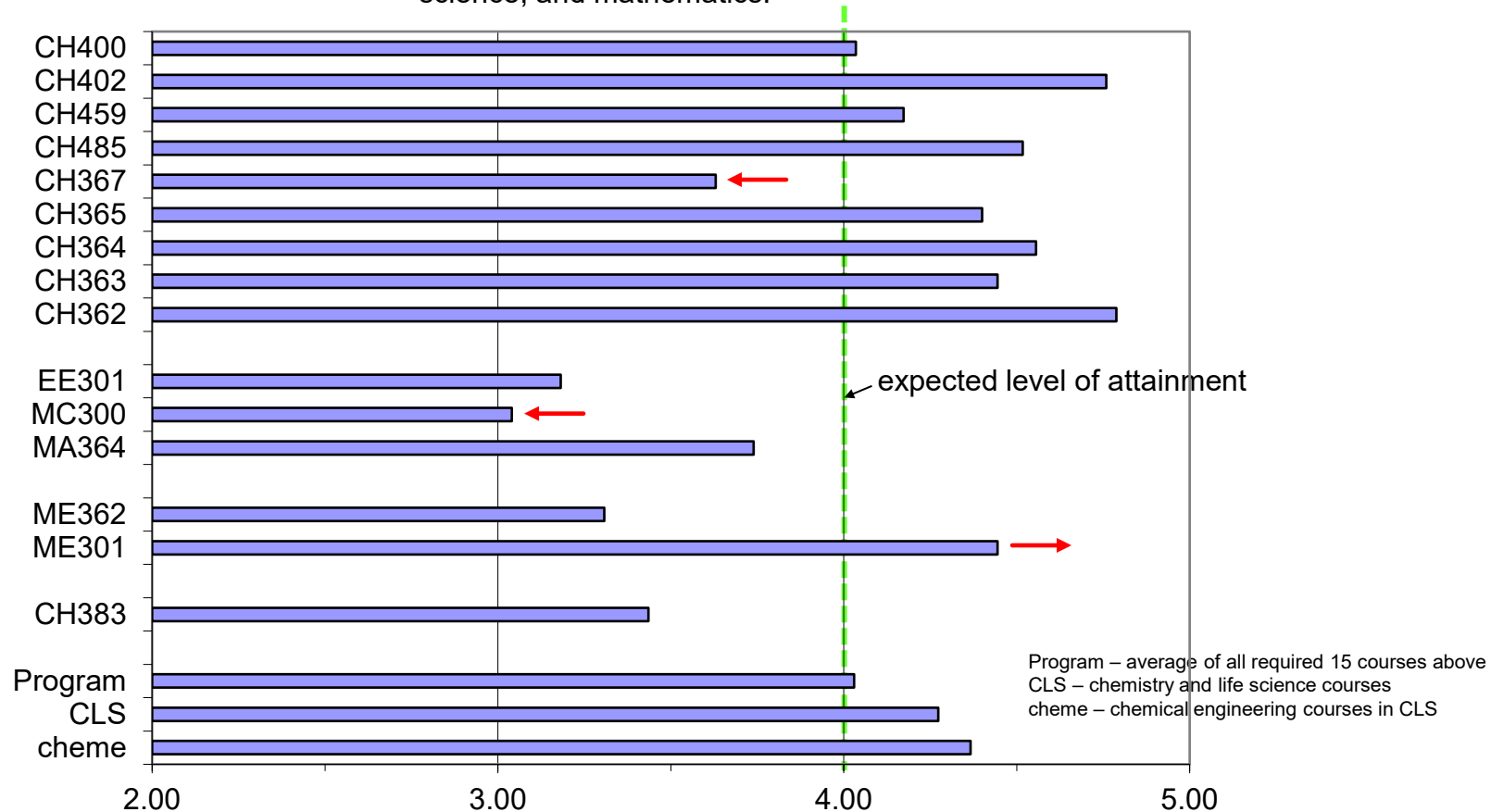
Expected levels of attainment are the 5-year averages



Example Data: End-of-Semester Surveys

Student Outcome 1

This course has improved my ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

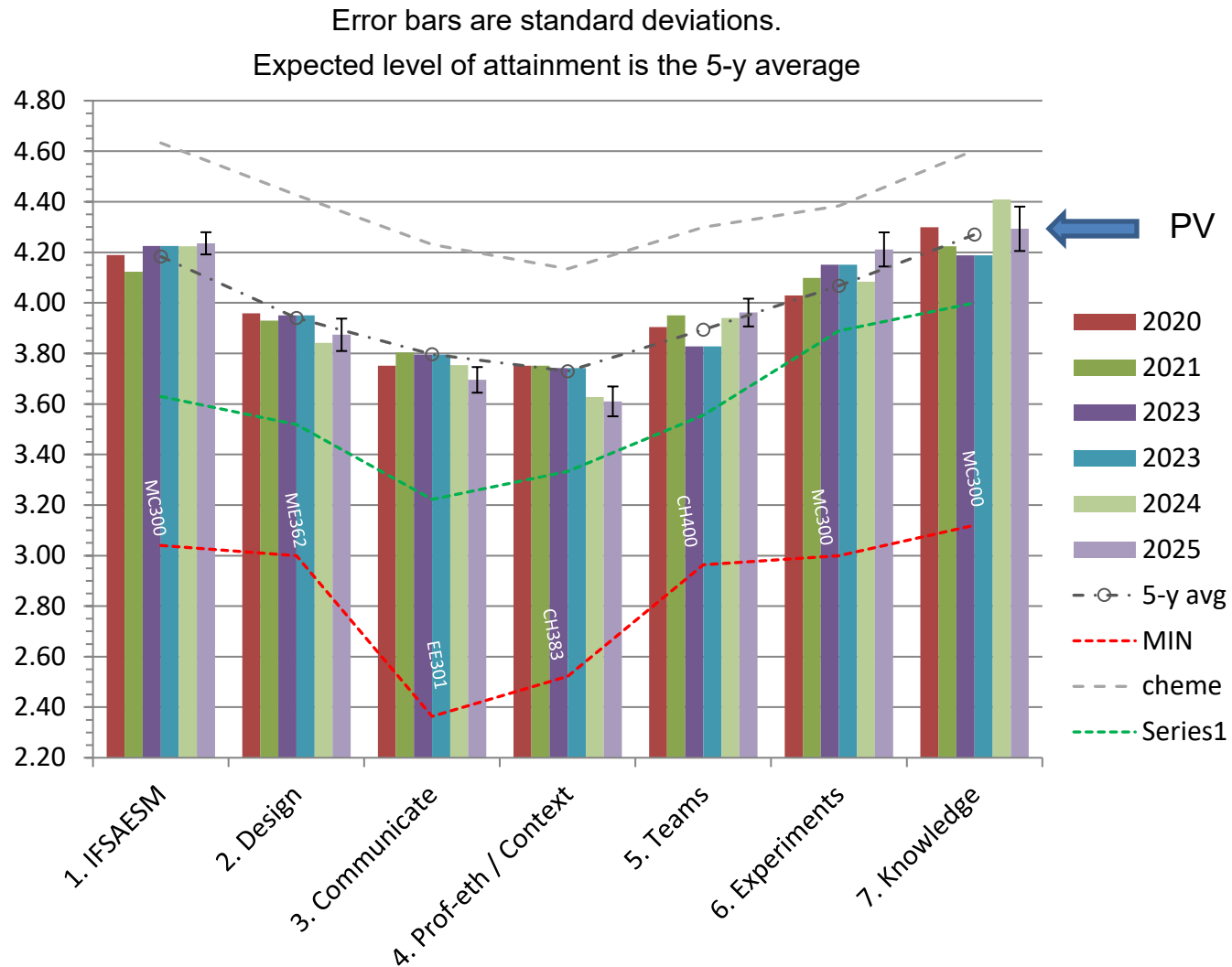


Data shown here is for Class of 2024
Similar data is collected for ABET student outcomes 1-7
Summary of all data is shown on next slide



End-of-Semester Surveys

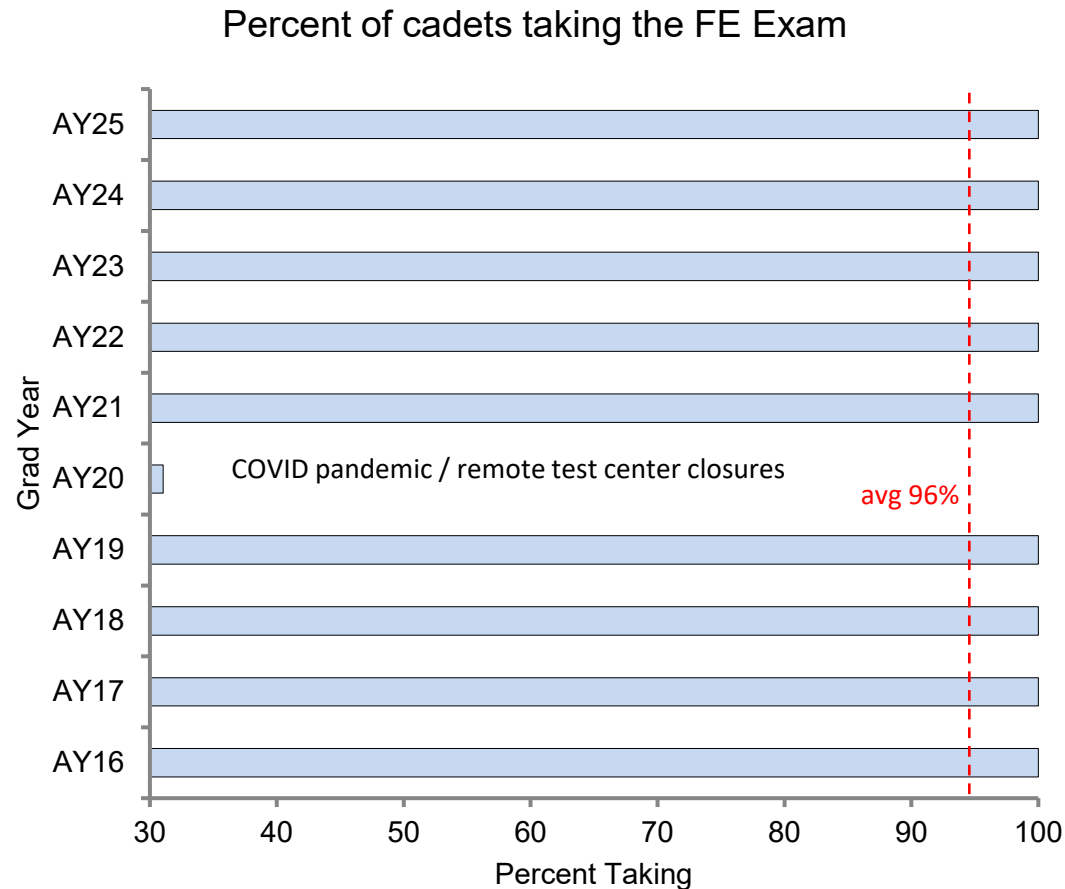
Program Averages from AY20-25





Fundamentals of Engineering Exam

Student Outcome 7: Acquire and apply new knowledge as needed, using appropriate learning strategies

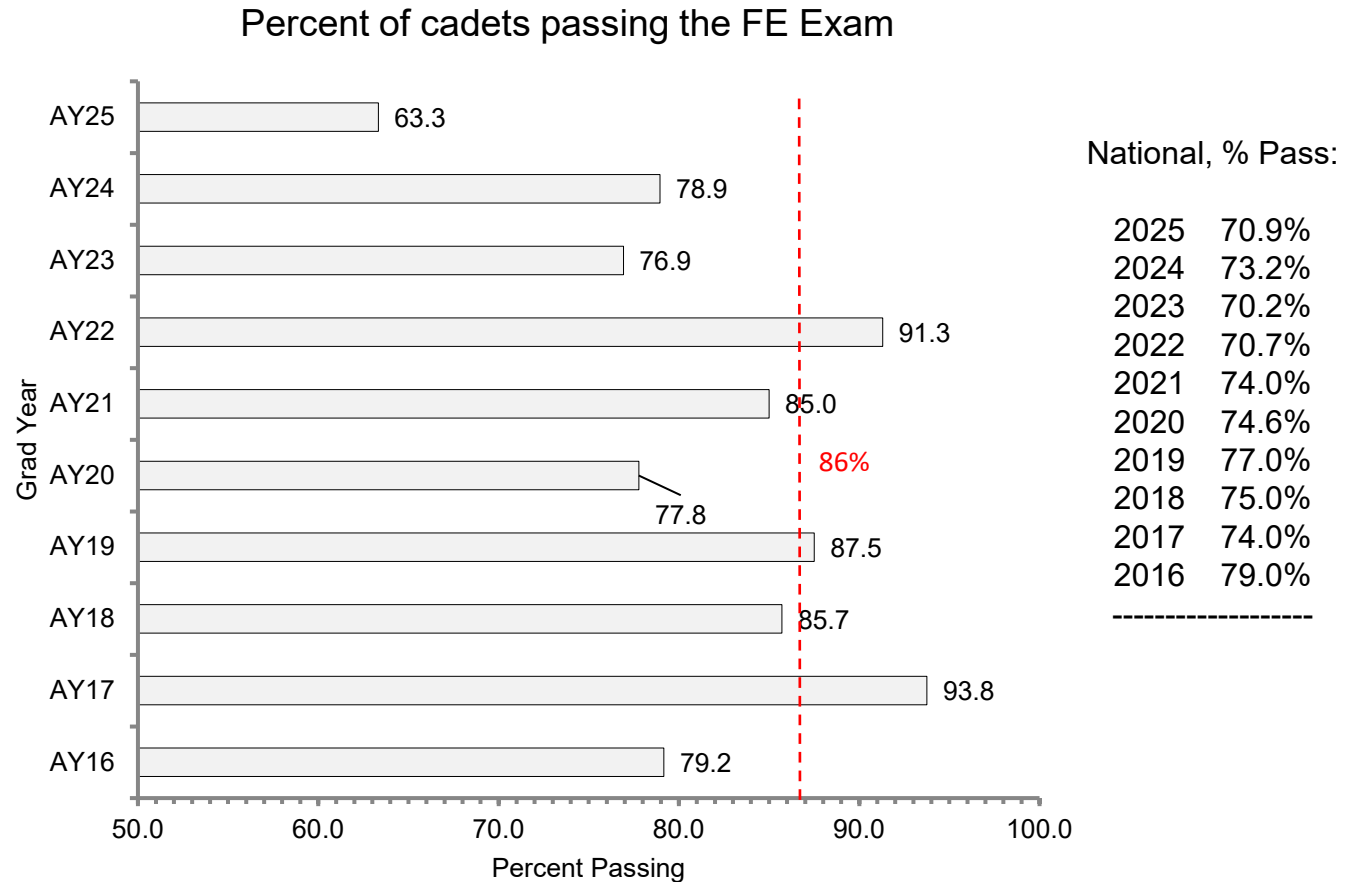


The average take rate was 86% prior to introduction of CH400 in AY2010.



Fundamentals of Engineering Exam

Student Outcome 1: Identify, formulate, and solve problems by applying principles of engineering, science, and mathematics.





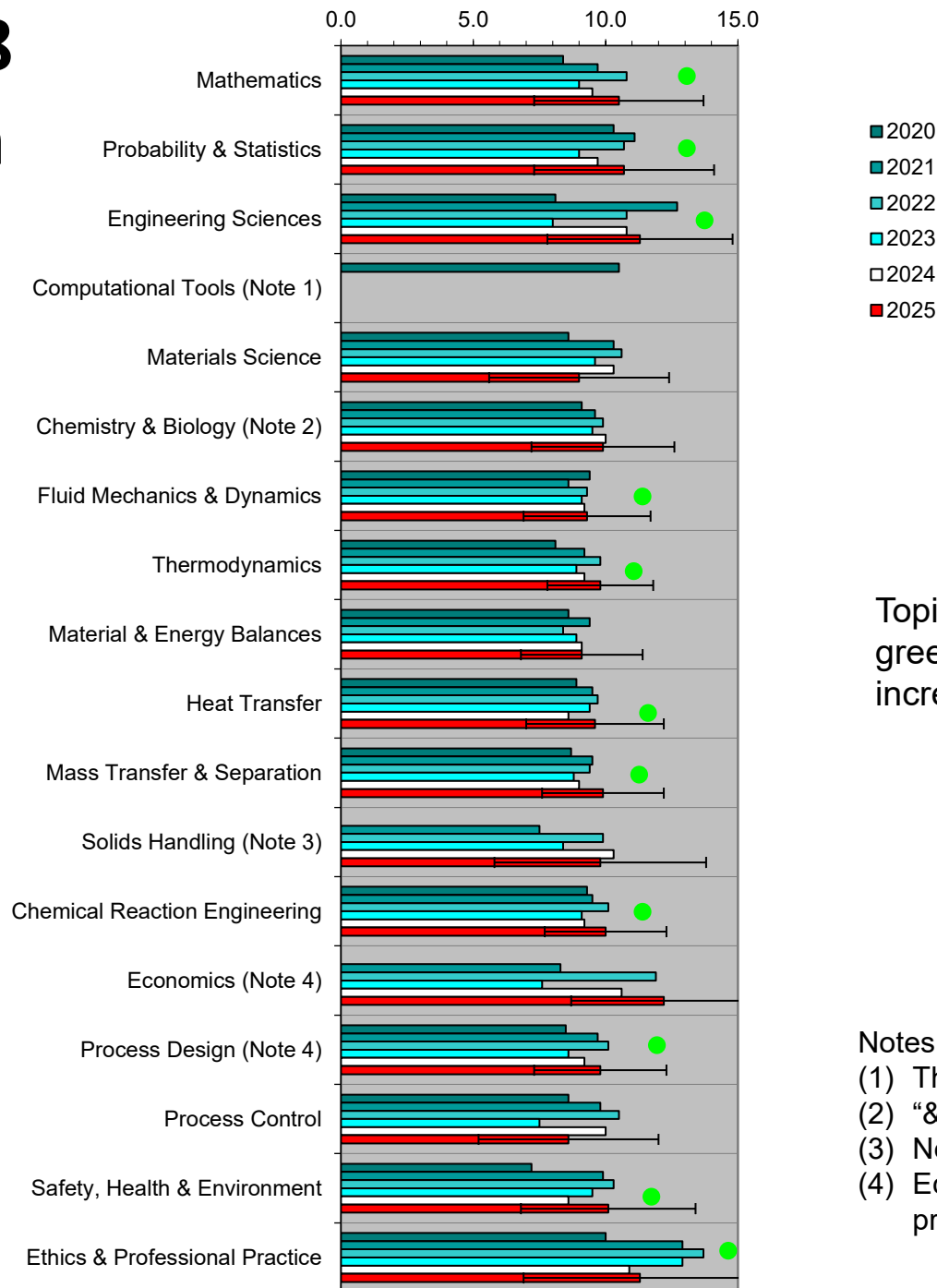
Outcome 8 Evaluation

FEE Results by Topic

AY20 to AY25

The error bars are
the individual
standard deviations
for the AY25 data.

The average
standard deviation
over all data is 3.0.



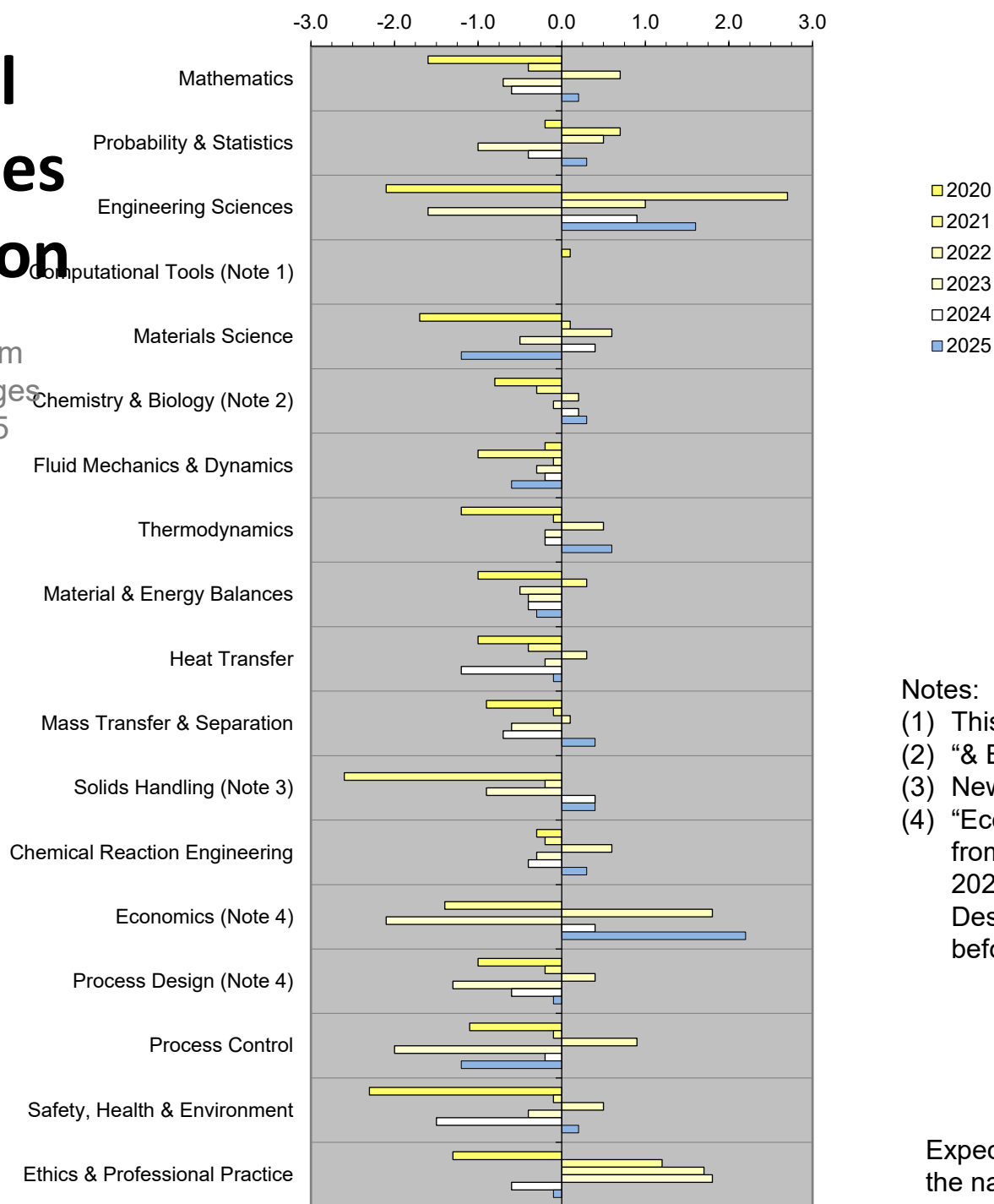
Notes:

- (1) This topic dropped in 2021.
- (2) "& Biology" added in 2021
- (3) New exam spec in 2021
- (4) Economics separated from process design in 2021.



Topical Outcomes Evaluation

Deviations from
National Averages
AY20 to AY25



Notes:

- (1) This topic dropped in 2021.
- (2) "& Biology" added in 2021
- (3) New exam spec in 2021
- (4) "Economics" separated from process design in 2021. Topic was Proc Design & Econ 2020 and before

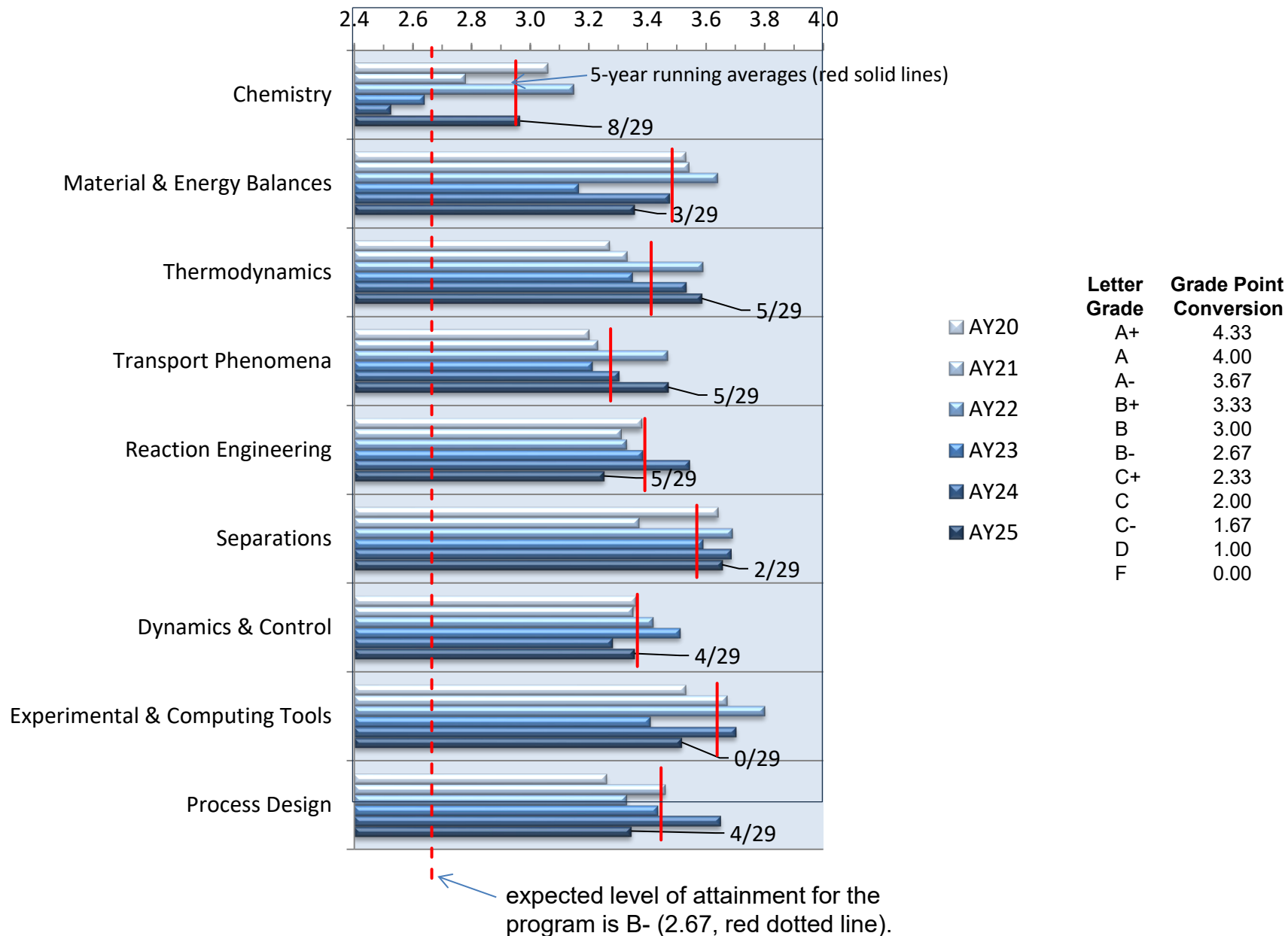
Expected level of attainment is the national average (0.0 line)



Topical Outcomes Evaluation

Student Outcome 8: Understanding of the Chemical Engineering Curriculum

Average GPA from Transcripts, AY2020 to AY2025



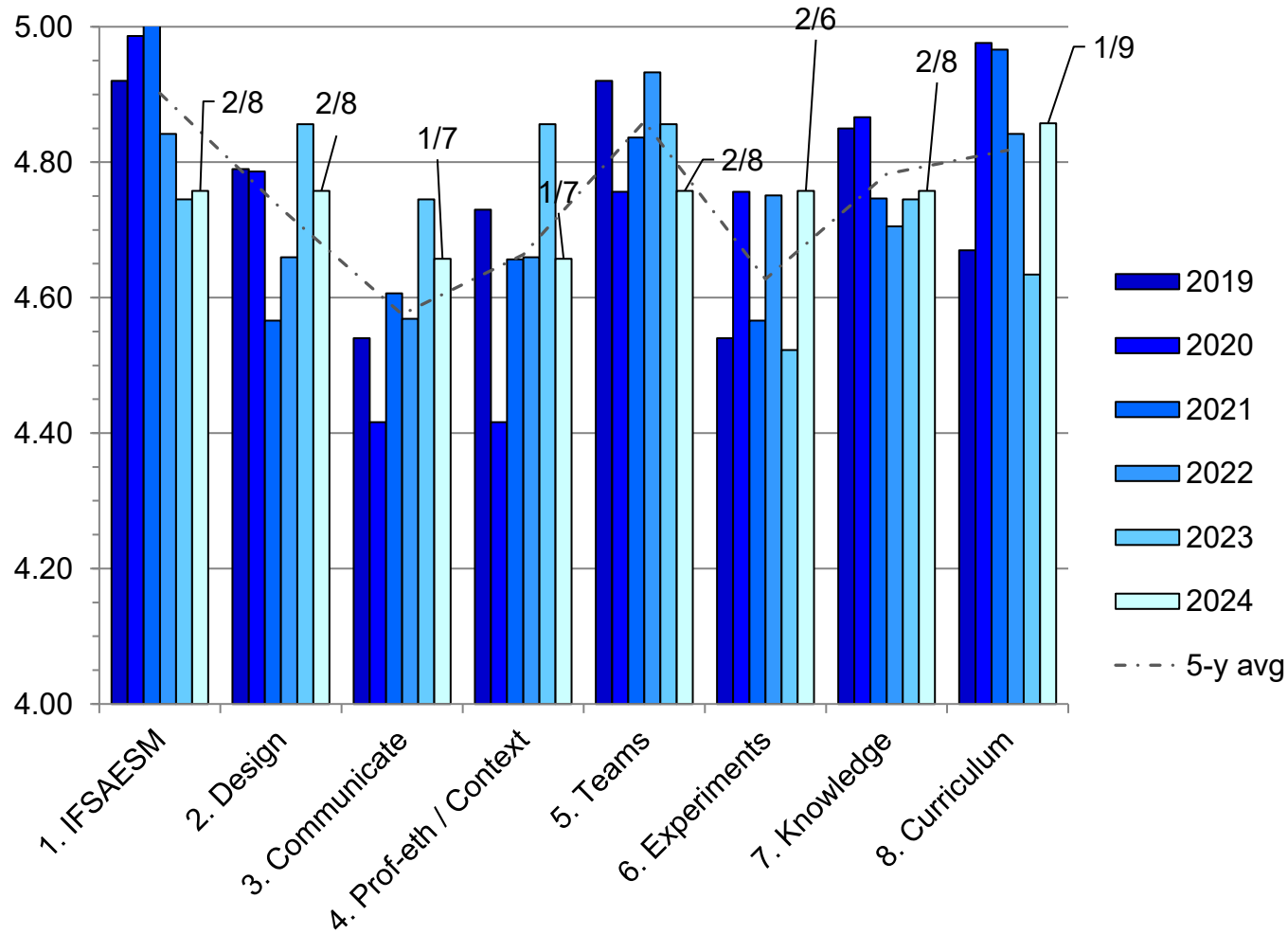


Advisory Board Student Outcomes Surveys

Student Outcomes 1-8

Program Averages from AY19-24

Data labels are response frequencies for 4 or 5 (# of 4s / # of 5s) on the 1-5 Survey Likert Scale
Standard deviations range from .00 to .52



AY25 data will be available after the next advisory board meeting in Spring 2026.